

# Widget Data Contract

One question per widget: what will the frontend ask you for, and what shape must the answer be?

HOW THE 37 TYPES ASK TODAY

3 send an aggregated query   34 ask for records

Every row in this document is read out of the code that builds the query - the same `queryFor` the running application calls - so nothing here is an intention. Read the publication contract first for what a Dataset must *declare*; this is about what gets *asked* afterwards.

## 1 · The request

One shape, for every widget:

```
interface DatasetQuery {  
  /** Group by these Dimensions. Omitted or empty means a single aggregate row. */  
  dimensions?: string[]  
  measures?: { field: string; aggregation: Aggregation }[]  
  timeRange?: { field: string; from?: string; to?: string; granularity?: TimeGranularity }  
  /** Keyed by Field key. Only Fields the Dataset declared `filterable`. */  
  filters?: Record<string, string | number>  
  sort?: { field: string; direction: 'ascending' | 'descending' }[]  
  limit?: number  
}  
  
type Aggregation = 'sum' | 'average' | 'count' | 'minimum' | 'maximum' | 'distinct-count'  
type TimeGranularity = 'day' | 'week' | 'month' | 'quarter' | 'year'
```

**Aggregation is asked for, never applied afterwards.** A widget requests "sum of revenue by region" and expects grouped, aggregated rows. It must not receive raw records and reduce them, for two reasons: it would let each widget decide what "sum" means, and it would ship records a Viewer may not be entitled to see.

An **aggregation** we send is always one the Dataset declared for that Measure. If we ask for something undeclared, that is our bug — reject it.

## 2 · The response

Four outcomes, and they must be distinguishable. This is the easiest part of the contract to get wrong, because nothing in the requirements says it about the API - only about the display.

```

type RetrievalOutcome =
  | { kind: 'rows'; rows: DatasetRow[]; totalCount: number }
  | { kind: 'empty' }      // authorized, and the Dataset has nothing to say
  | { kind: 'denied' }     // not authorized for this Dataset
  | { kind: 'withdrawn' }  // the Dataset is gone

type DatasetRow = Record<string, string | number | null>

```

A response of `[]` cannot carry this: empty, denied and withdrawn all look identical, and the frontend would have to guess. It renders each of the four differently — a denial says access was denied rather than showing zero, because showing zero teaches a Viewer the figure *is* zero.

**A genuine failure is an error, not an outcome** — the absence of an answer rather than one of the answers.

`kind: 'rows'` with an empty `rows` array is treated as a **contract breach** and surfaced as a failure, not as **empty**. Send `{ kind: 'empty' }`.

### 3 · What a row must look like

Your integration guide says the relayed body's *"shape must match frontend visualization requirements"* and then, reasonably, does not say what those are. This section is that half of the sentence.

**Field names are yours, not ours.** This is the first thing to settle, because it is the thing teams most often expect us to dictate. We never need a field called `value`. Every visualisation takes its field references as configuration — a bar chart is handed `valueKey: 'settlement_value'`, and the name is bound once when an Author builds the Widget. So `val`, `value`, `total_amount` and `settlement_value` are all equally fine, and asking you to rename one would break every existing binding for no gain. What we need is that a declared name **matches the key you return** — which your guide already requires.

What we do need is below. Each is an invariant rather than a preference, and each names the failure it avoids, because every one of these fails *silently*.

#### Values carry their declared type

A Field declared `number` must arrive as a JSON number.

```

{ "region": "EMEA", "revenue": 1234.5 }    // yes
{ "region": "EMEA", "revenue": "1,234.50" } // no

```

A formatted string becomes `NaN` the moment we coerce it, and `NaN` in an SVG path renders as **nothing at all** — no error, no warning, an empty chart that looks like missing data. Our own test suite cannot catch this, because it tests our fixtures and those are well-formed by construction. Same for `boolean`: `true`, not `"true"` and not `1`.

#### Dates are ISO-8601, and one grain per Dataset

2026-08-07 for a day, 2026-08 for a month, 2026 for a year. Not 07/08/2026, not a locale string, not epoch milliseconds.

Date filtering compares these as strings, which is exact and fast for ISO-8601 and wrong for everything else. 07/08/2026 would not throw — it would return the wrong rows, quietly, which is worse. Mixing grains inside one Dataset breaks the same comparison.

### Absent means null, not a missing key and not ""

```
{ "region": "EMEA", "revenue": null } // yes — no value
{ "region": "EMEA" } // no — key omitted
{ "region": "EMEA", "revenue": "" } // no — empty string
```

An omitted key and an explicit null take different paths through several visualisations, and "" coerces to 0 in a Measure — which draws a bar of height zero where there should be a gap. A zero and a nothing are different claims about the world.

### One row grain, and say what it is

A row should mean one thing for the whole Dataset: one row per day, or per region, or per transaction. Section 5 lists the volume each widget can usefully draw — a calendar heatmap wants about 365 rows, a stat card wants one — and that is guidance, not a limit for you to enforce.

The thing to avoid is a Dataset whose grain shifts with its filters, because a widget bound to it is correct on one query and meaningless on the next.

### Order, or tell us there is none

Nine of the 37 widget types need rows in order — every trend, the chronological pair, and the temporal ones. A line chart drawn from unordered rows is not untidy, it is wrong: the line doubles back on itself.

Since the query carries no sort, we need one of two answers per Dataset. Either the endpoint returns a stable order and says so, or it does not and we sort in the browser. Both are workable; not knowing which is not.

### A location Field should say which kind it is

FieldType: 'location' covers two different things, and they feed different visualisations. A region name ("Kenya") shades a choropleth; a coordinate (-1.29) places a point on a map, and a point needs two Fields that know which of them is latitude. Today nothing distinguishes them, so a point map can be offered a table of regional sales and plot revenue as a latitude. We have raised this upstream; until it is settled, please say in the Field's description which one you mean.

## 4 · Aggregation grain - the thing to settle with us

Of the 37 built widget types, **3 currently send an aggregated query and 34 ask for records**. That is not a recommendation, it is a report - and it needs a conversation before it meets a real database.

That is not a recommendation, it is a report — and it needs a conversation before it meets a real database.

The reason is our fixtures: most are already stored at the grain the chart draws. `sales-by-region` is one row per region, so a bar chart over it asks for records and gets exactly the six bars it wants. Against a raw table the same widget would pull every transaction and group in the browser, which is both slow and the thing §2 of the publication contract forbids.

#### What this means for you:

- The 3 single-aggregate types are correct as they stand and safe against any volume.
- For the rest, tell us where your data's grain sits. If a Dataset is already aggregated to the grain we draw, records are fine and the `rows` count in §5 is what to expect. If it is raw, we need to send `dimensions + measures` and we will fix the query — the mapping already carries the information, so it is our change, not yours.
- The row counts in §5 are what our *fixtures* return. Treat them as the volume a widget can usefully draw, not as a limit you should enforce: a calendar heatmap wants 365 points and a stat card wants 1.

This is tracked on our side as divergences **D13** and **D14** in `DIVERGENCES.md`.

## 5 · Every widget, and what it asks for

**Needs** is what a Dataset must offer for the widget to be *offered* at all - the Field roles, and how many of each. A widget whose required slots cannot be filled is never shown to an Author, so a Dataset missing a Time Dimension simply does not produce trend charts.

**Rows** is what our fixtures return: the volume a widget can usefully draw, not a limit to enforce. A calendar heatmap wants 365 points and a stat card wants one.

Single value single-value

| TYPE                                                | NEEDS                                                | REQUEST TODAY    | GRAIN            | ROWS |
|-----------------------------------------------------|------------------------------------------------------|------------------|------------------|------|
| stat-card<br><small>Stat card</small>               | Measure — 1 × Measure                                | sum(revenue)     | SINGLE AGGREGATE | 1    |
| sparkline-card<br><small>Sparkline card</small>     | Measure — 1 × Measure<br>Period — 1 × Time Dimension | order by date ↑  | RECORDS          | 365  |
| delta-card<br><small>Delta card</small>             | Measure — 1 × Measure<br>Period — 1 × Time Dimension | order by month ↑ | RECORDS          | 24   |
| progress-tracker<br><small>Progress tracker</small> | Actual — 1 × Measure<br>Target — 1 × Measure         | order by month ↑ | RECORDS          | 24   |

Trend trend

| TYPE                                        | NEEDS                                                      | REQUEST TODAY    | GRAIN   | ROWS |
|---------------------------------------------|------------------------------------------------------------|------------------|---------|------|
| line-chart<br><small>Line chart</small>     | Time axis — 1 × Time Dimension<br>Measures — 1–5 × Measure | order by month ↑ | RECORDS | 24   |
| area-chart<br><small>Area chart</small>     | Time axis — 1 × Time Dimension<br>Measures — 1–5 × Measure | order by date ↑  | RECORDS | 365  |
| spline-chart<br><small>Spline chart</small> | Time axis — 1 × Time Dimension<br>Measures — 1–5 × Measure | order by month ↑ | RECORDS | 24   |
| step-chart<br><small>Step chart</small>     | Time axis — 1 × Time Dimension<br>Measures — 1–5 × Measure | order by month ↑ | RECORDS | 24   |

Categorical comparison categorical-comparison

| TYPE                                   | NEEDS                                                                                  | REQUEST TODAY                | GRAIN   | ROWS |
|----------------------------------------|----------------------------------------------------------------------------------------|------------------------------|---------|------|
| bar-chart-vertical<br>Bar chart        | <b>Categories</b> — 1 × Dimension or Time Dimension<br><b>Measures</b> — 1–4 × Measure | no aggregation — all records | RECORDS | 6    |
| bar-chart-horizontal<br>Horizontal bar | <b>Categories</b> — 1 × Dimension or Time Dimension<br><b>Measures</b> — 1–4 × Measure | no aggregation — all records | RECORDS | 12   |
| grouped-bar-chart<br>Grouped bar       | <b>Categories</b> — 1 × Dimension or Time Dimension<br><b>Measures</b> — 2–4 × Measure | no aggregation — all records | RECORDS | 6    |
| stacked-bar-chart<br>Stacked bar       | <b>Categories</b> — 1 × Dimension or Time Dimension<br><b>Measures</b> — 2–4 × Measure | no aggregation — all records | RECORDS | 6    |

Composition composition

| TYPE                       | NEEDS                                                             | REQUEST TODAY                | GRAIN   | ROWS |
|----------------------------|-------------------------------------------------------------------|------------------------------|---------|------|
| pie-chart<br>Pie chart     | <b>Segments</b> — 1 × Dimension<br><b>Measure</b> — 1 × Measure   | no aggregation — all records | RECORDS | 6    |
| donut-chart<br>Donut chart | <b>Segments</b> — 1 × Dimension<br><b>Measure</b> — 1 × Measure   | no aggregation — all records | RECORDS | 6    |
| treemap<br>Treemap         | <b>Rectangles</b> — 1 × Dimension<br><b>Measure</b> — 1 × Measure | no aggregation — all records | RECORDS | 12   |

Ranking & flow ranking-and-flow

| TYPE                       | NEEDS                                                                                   | REQUEST TODAY                | GRAIN   | ROWS |
|----------------------------|-----------------------------------------------------------------------------------------|------------------------------|---------|------|
| ranked-list<br>Ranked list | <b>Entries</b> — 1 × Dimension<br><b>Measure</b> — 1 × Measure                          | no aggregation — all records | RECORDS | 20   |
| leaderboard<br>Leaderboard | <b>Entries</b> — 1 × Dimension<br><b>Measure</b> — 1 × Measure                          | no aggregation — all records | RECORDS | 12   |
| funnel<br>Funnel           | <b>Stages</b> — 1 × Dimension<br><b>Measure</b> — 1 × Measure                           | no aggregation — all records | RECORDS | 6    |
| sankey<br>Sankey           | <b>From</b> — 1 × Dimension<br><b>To</b> — 1 × Dimension<br><b>Volume</b> — 1 × Measure | no aggregation — all records | RECORDS | 10   |

Status status

| TYPE                                       | NEEDS                                                                                                 | REQUEST TODAY                | GRAIN            | ROWS |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------|------------------------------|------------------|------|
| status-list<br>Status list                 | <b>Entities</b> — 1 × Dimension<br><b>State</b> — 1 × Dimension<br><b>Measure</b> — up to 1 × Measure | no aggregation — all records | RECORDS          | 9    |
| status-indicator<br>Status tile            | <b>Entities</b> — 1 × Dimension<br><b>State</b> — 1 × Dimension                                       | no aggregation — all records | RECORDS          | 9    |
| threshold-indicator<br>Threshold indicator | <b>Measure</b> — 1 × Measure                                                                          | average(uptime)              | SINGLE AGGREGATE | 1    |
| alert-banner<br>Alert banner               | <b>Measure</b> — 1 × Measure                                                                          | average(errorRate)           | SINGLE AGGREGATE | 1    |

Radial radial

| TYPE                       | NEEDS                                                              | REQUEST TODAY                | GRAIN   | ROWS |
|----------------------------|--------------------------------------------------------------------|------------------------------|---------|------|
| gauge<br>Gauge             | <b>Actual</b> — 1 × Measure<br><b>Target</b> — 1 × Measure         | order by month ↑             | RECORDS | 24   |
| radar-chart<br>Radar chart | <b>Entities</b> — 1 × Dimension<br><b>Measures</b> — 3–8 × Measure | no aggregation — all records | RECORDS | 12   |

Tabular tabular

| TYPE                       | NEEDS                                                                                               | REQUEST TODAY                | GRAIN   | ROWS |
|----------------------------|-----------------------------------------------------------------------------------------------------|------------------------------|---------|------|
| data-table<br>Data table   | <b>Columns</b> — 1–12 × Dimension or Time Dimension or Measure                                      | no aggregation — all records | RECORDS | 120  |
| pivot-table<br>Pivot table | <b>Rows</b> — 1 × Dimension<br><b>Columns</b> — 1 × Dimension<br><b>Measure</b> — up to 1 × Measure | no aggregation — all records | RECORDS | 120  |

**Distribution** distribution

| TYPE                   | NEEDS                                                         | REQUEST TODAY                | GRAIN   | ROWS |
|------------------------|---------------------------------------------------------------|------------------------------|---------|------|
| histogram<br>Histogram | <b>Measure</b> — 1 × Measure                                  | no aggregation — all records | RECORDS | 2000 |
| box-plot<br>Box plot   | <b>Groups</b> — 1 × Dimension<br><b>Measure</b> — 1 × Measure | no aggregation — all records | RECORDS | 2000 |

**Correlation** correlation

| TYPE                         | NEEDS                         | REQUEST TODAY                | GRAIN   | ROWS |
|------------------------------|-------------------------------|------------------------------|---------|------|
| scatter-plot<br>Scatter plot | <b>Measures</b> — 2 × Measure | no aggregation — all records | RECORDS | 12   |
| bubble-chart<br>Bubble chart | <b>Measures</b> — 3 × Measure | no aggregation — all records | RECORDS | 12   |

**Temporal pattern** temporal-pattern

| TYPE                                 | NEEDS                                                                                                                                             | REQUEST TODAY                | GRAIN   | ROWS |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|---------|------|
| calendar-heatmap<br>Calendar heatmap | <b>Date</b> — 1 × Time Dimension<br><b>Measure</b> — 1 × Measure                                                                                  | order by date ↑              | RECORDS | 365  |
| cohort-grid<br>Cohort grid           | <b>Cohorts</b> — 1 × Time Dimension<br><b>Elapsed period</b> — 1 × Dimension or Time Dimension<br><b>Measure</b> — 1 × Measure                    | no aggregation — all records | RECORDS | 78   |
| timeline-chart<br>Gantt chart        | <b>Rows</b> — 1 × Dimension<br><b>Start and end</b> — 2 × Measure<br><b>Grouping</b> — up to 1 × Dimension<br><b>Progress</b> — up to 1 × Measure | no aggregation — all records | RECORDS | 9    |

**Chronological** chronological

| TYPE                                          | NEEDS                                                                                             | REQUEST TODAY                | GRAIN   | ROWS |
|-----------------------------------------------|---------------------------------------------------------------------------------------------------|------------------------------|---------|------|
| activity-feed<br><small>Activity feed</small> | <b>When</b> — 1 × Time Dimension<br><b>Actor</b> — 1 × Dimension<br><b>Action</b> — 1 × Dimension | no aggregation — all records | RECORDS | 60   |
| event-log-view<br><small>Event log</small>    | <b>When</b> — 1 × Time Dimension<br><b>Columns</b> — 1–8 × Dimension or Time Dimension or Measure | no aggregation — all records | RECORDS | 60   |

**Geospatial** geospatial

| TYPE                                  | NEEDS                                                                                                                                                               | REQUEST TODAY                | GRAIN   | ROWS |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|---------|------|
| point-map<br><small>Point map</small> | <b>Place</b> — 1 × Dimension, geographic<br><b>Latitude</b> — 1 × Measure, geographic<br><b>Longitude</b> — 1 × Measure, geographic<br><b>Measure</b> — 1 × Measure | no aggregation — all records | RECORDS | 20   |

**6 · Field roles, for reference**

| ROLE           | WHAT IT IS                                  | NOTES                                          |
|----------------|---------------------------------------------|------------------------------------------------|
| dimension      | Identifies or categorizes a record          |                                                |
| time-dimension | A Dimension whose values are points in time | Counts as a Dimension wherever one is required |
| measure        | Can be meaningfully aggregated              | Must declare which aggregations are meaningful |

Two things a Dataset declares that change what we can offer:

- **filterable gates every filter.** A Viewer-facing filter or a Dashboard Control can only act on a Field you marked filterable. We enforce this twice — the picker will not offer it, and the query builder drops it — so an unmarked Field is unreachable by design.
- **semantic unlocks four Families.** Role says what a Field *is* and cannot say what it is *for*. A point map needs to know which Measure is a latitude; without it, a ranked list will happily rank countries by how far north they are. See `PUBLICATION_CONTRACT.md` for the values.

**7 · What we will not ask you for**

- **Writes.** Every widget is read-only, enforced by the props carrying no callback — there is nothing a widget *could* call to mutate.
- **Joins.** A widget draws from exactly one Dataset. If two Datasets need relating, that is a Dataset the Source System publishes, not a query we send.
- **Formatting.** Send numbers as numbers. `format` on a Field tells us how to render them.
- **Anything a Viewer may not see.** Authorization is resolved per Dataset per call, on your side. We ask; we do not carry a rule about who may read what.